DV1566 H24 LP2 INTRODUKTION TILL CLOUD COMPUTING (LABORATORY 1 – CONTAINERIZATION)

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2.1 QUESTIONNAIRE

2.1.1 Getting Started

Q1: Why do you run the container docker/getting-started in detached mode?
Answer: Running a container in detached mode allows the container to run in the background without tying up the terminal. This is useful for services that need to keep running while performing other tasks.

Q2: What is the difference between a container and a container image?
Answer: A container image defines the environment and application, while a container is a running instance of that particular container image.

2.1.2 Sample Application

Q1: What is the meaning of the Docker directives used in the Dockerfile? Please comment on each of the directives.
Answer: A Dockerfile is a script containing a series of instructions on how to build a Docker image and each derivative has its own specific functionalities. Those are given below:

a. FROM: Specifies the base image to use for the container.

b. RUN: Executes commands to install packages or set up the environment.

c. CMD: Defines the default command to run when the container starts.

d. COPY: Copies files from the host to the container.

e. EXPOSE: Informs Docker about the ports the container will listen on.

Q2: Why is it important to tag a container image?
Answer: Tagging helps to identify and manage specific versions.

Q3: Why should we bind a host port with the container port?
Answer: Binding a host port to a container port allows external access to the containerized application.

2.1.3 Update the Application

Q1: Is it possible to bind two containers on the same host port?
Answer: No, two containers cannot be bound at the same host port simultaneously due to conflicts.

Q2: Why, after stopping a container, do you need to remove it?
Answer: Stopped containers still occupy disk space. Removing them frees up resources.

Q3: Is it possible to remove a running container without stopping it first?
Answer: No, the container must be removed before stopping it, otherwise Docker will throw an error.

2.1.4 Share the Application

Q1: Given a container image available on a Docker image repository, can you start an instance of the image on any Docker host? Are there any limitations?
Answer: Yes, initiation of an instance of a container image on any Docker host can be done. However, it is essential to ensure compatibility between the host environment and the image to guarantee optimal performance.

2.1.5 Persist the DB

Q1: If you run two instances of the same container image (A and B) and create a file in container A, is that new file visible in container B?
Answer: No, files created in one container are not visible in another unless they share the same volume.

Q2: What can you do with the docker exec command?
Answer: The docker exec command allows the user to run commands inside a running container.

Q3: Why do you need to mount the volume in the container filesystem? Does this modify the container filesystem?
Answer: Mounting a volume helps keep the data safe even after the container is stopped or removed. Instead of changing the container's filesystem, it connects external storage to the container, so that the data remains intact and accessible.

Q4: Where is the volume located in the file system of your Docker host?
Answer: Volumes are located in the /var/lib/docker/volumes directory on my Docker host.

2.1.6 Use Bind Mounts

Q1: What is a development container?
Answer: A development container is a lightweight environment that includes all the tools and dependencies needed for software development and it is used for coding and testing.

Q2: When you are in development mode, do you need to rebuild the container each time you update the app to see the changes? Why?
Answer: No, bind mounts enable live updates in the container without rebuilding.

2.1.7 Multi-Container App & Docker Compose

Q1: What is a Docker service?
Answer: A Docker service is a container managed by Docker Swarm or Kubernetes for scaling and orchestration.

Q2: Can we spin up a single instance of a Docker container using a Docker Compose file?
Answer: Yes, Docker Compose can spin up a single container.

Q3: Can we run a MySQL container and store the database structure and data in a volume?
Answer: Yes, MySQL data can be stored in a volume to ensure persistence.

Q4: Is the order of the services defined in a Docker Compose file important, or is it irrelevant which service is defined first?
Answer: The order of services is not mandatory, but logical dependencies may require certain services to start first.

Q5: Is it mandatory to define the network in a Docker Compose file?
Answer: No, it is not mandatory to define the network in a Docker Compose file because Docker Compose file automatically creates a network unless defined.

Q6: If you would like to run a multi-container app, is it necessary to use Docker Compose (i.e., to define a service), or can you achieve the same objective using Docker commands from the shell? Does a service offer more than just running a multiple-container app with a single command?
Answer: No, using Docker commands can achieve the same objective. However, Docker Compose simplifies management with a single command and additional configuration options.